

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (Biotechnology) (Credit System) Degree Examinations

January 2023

19BT701 / 17BT701 – IMMUNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Briefly comment on i) Neutrophil and ii) Macrophage.	10	L*1	1	1,2
	b) Discuss on defensive mechanism of adoptive immunity.	10	L1	1	1,2
2.	a) Describe the alternate pathway of complement system and its role in immune response.	10	L1	1	1,2
	b) How vaccines are evolved with different generations? Explain.	10	L2	1	1,2
Unit – II					
3.	a) Narrate the fine structure of antibody with neat labelled diagram	10	L1	2	1,3
	b) Discuss antigen process and presentation with the help of endocytic pathway.	10	L2	2	1,3
4.	a) Describe the structure of class II MHC molecules with its significances.	10	L1	2	1,3
	b) Comment on Isotypes, allotypes and idiotypes.	10	L1	2	1,3
Unit – III					
5.	a) Elaborate on proliferation and differentiation of T-cells and add a note on its significances.	10	L1	3	1,3
	b) Write a short note on cytokines and its significances.	10	L1	3	1,3
6.	a) With reference to B cell explain the following i) Origin ii) Development iii) Activation and iv) Selection	10	L1	3	1,3
	b) Illustrate the structural features of the immunoglobulin which can pass through placenta.	10	L2	3	1,3
Unit – IV					
7.	a) Comment on organ specific autoimmune disorder with suitable example of disease manifestation.	10	L2	4	1,2
	b) Allergens contact leads to hypersensitivity reactions in some individuals. Elaborate with suitable example.	10	L2	4	1,2
8.	a) Give a brief account on formation of granulomata and add a note on hypersensitivity reaction responsible for this.	10	L1	4	1,2
	b) Write the mechanism of cell mediated hypersensitivity reaction.	10	L2	4	1,2
Unit – V					
9.	a) Illustrate the role of T – cell in transplant rejection.	10	L1	5	1,4
	b) Explain immune tolerance using allophenic mice as an example.	10	L2	5	1,4
10.	a) Can monoclonal antibodies prevent graft rejection? Elaborate.	10	L2	5	1,4
	b) How tumors evade immune response? Explain the process.	10	L2	5	1,4

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
